

# ZTA Design Clinic Workbook

This is your team's answer sheet. Work through one use case at a time: sketch your design, explain how it solves the customer's problems, and list the Cisco products you would use. Your proctor scores each use case out of 100 and gives you feedback.

## How to use this workbook

- 1 Pick your roles.** As a group, assign the four roles (NetOps, Information Security, Compliance and Scribe). The Scribe records the team's answers in this workbook.
- 2 Understand the customer.** Review the current customer architecture and, for each use case, its requirements and pain points. The key points are repeated in this workbook so you can answer them directly.
- 3 Design your solution.** Agree on a Cisco Zero Trust design that meets every requirement and solves each pain point. Do use cases 1-3 first (required); 4-5 are bonus if time permits.
- 4 Draw your diagram.** Sketch your proposed network in the diagram box, or paste a diagram you made with the Cisco Circuit app or another AI tool.
- 5 Write your responses.** In the tables, explain how your design addresses each pain point and requirement. Then list each Cisco product, what it does, and why you chose it. Note any Cisco differentiator.
- 6 Present to your proctor.** Be ready to walk your SME / proctor through your design and explain your choices. They score each use case out of 100 and give you feedback.

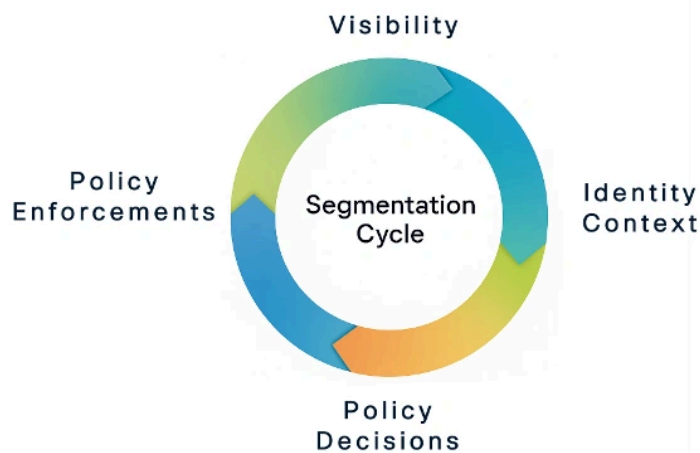
**This is a 60-minute activity.** Do use cases 1-3 (required) first; 4-5 are stretch goals if you finish early. Keep **one diagram** and grow it as you go, and answer in **short bullets**.

- Use cases 1-3 are the deliverable; 4-5 are stretch goals if you finish early.
- Keep one diagram and grow it use case to use case, don't start over each time.
- Answer in short bullet points, not paragraphs. Name the product, say what it does, say why.
- Ask your proctor for a 60-second sanity check after Use Case 1 so you're on the right track.

**How your work is scored.** Each use case is worth **100 points**, **Mapping 40** (how well you address the pain points & requirements) · **Products 30** (the right Cisco products, explained clearly) · **Diagram 30**. Your group earns one **cumulative level (Associate · Professional · Expert)** from the total across use cases: complete the three required use cases for **Associate** or, at 70% or higher, **Professional**; attempt all five and reach 80% or higher for **Expert**. See the **Final score & group level** page at the end for the exact rules. New to a product name? See the **Glossary** and the product primer in your Activity Document.

# The Segmentation Cycle

A practical approach to network security. Apply this method in every use case.



Team / table #: \_\_\_\_\_ Members: \_\_\_\_\_

## Use Case #1: Managed Environment; Segmentation

Foundation · Required

100 pts

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

### STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here, or paste one made with the Cisco Circuit app / an AI tool.

## STEP 2 · ADDRESS THE CUSTOMER'S PAIN POINTS

| CUSTOMER PAIN POINT                                                                        | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Prior segmentation failed. Using VLANs/IP subnets is complex and tedious.                  |                                                         |
| Profiling is labor intensive and staff unable to keep up with constant new devices joining |                                                         |
| Failing PCI audits                                                                         |                                                         |
| Corporate PCs not running latest AntiVirus and OS updates                                  |                                                         |
| No budget for buying more firewalls to do segmentation                                     |                                                         |

## STEP 3 · MEET THE CUSTOMER'S REQUIREMENTS

| CUSTOMER REQUIREMENT                                                                    | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------|
| Employees / IoT / deskagents (Mac-minis) / Guest groups                                 |                                                         |
| IoT should be restricted from the other groups but need data center and Internet access |                                                         |
| Guests should have Internet access only                                                 |                                                         |
| Employee and IoT require controlled internet, and data center application access        |                                                         |
| Deskside Mac-minis have access to on-prem data center only                              |                                                         |

## STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled. Explain **how each works** and **why you'd choose it** (the differentiator). Use the blank row for anything else you'd add.

| CISCO PRODUCT / FEATURE                                | WHAT IT DOES / HOW IT WORKS | WHY YOU CHOSE IT (CISCO DIFFERENTIATOR) |
|--------------------------------------------------------|-----------------------------|-----------------------------------------|
| ISE                                                    |                             |                                         |
| ISE-SGT                                                |                             |                                         |
| ISE SGT ACL policy                                     |                             |                                         |
| ISE pxGrid                                             |                             |                                         |
| Catalyst SD-WAN                                        |                             |                                         |
| FMC / SCC / FTD                                        |                             |                                         |
| FMC / SCC / FTD,<br>Source/Dest SGT policy enforcement |                             |                                         |
| Cisco Secure Workload,<br>App Dependency Mapping       |                             |                                         |
| Cisco Secure Workload,<br>pxGrid integration with ISE  |                             |                                         |
|                                                        |                             |                                         |

PROCTOR USE ONLY: USE CASE SCORE

| CRITERION                    | MAX | SCORE |
|------------------------------|-----|-------|
| Pains & requirements mapping | 40  |       |
| Cisco products (how & why)   | 30  |       |
| Diagram                      | 30  |       |
| Use case total               | 100 |       |

Carry this use case total to the **Final score & group level** page. The Associate / Professional / Expert level and written feedback are recorded once there, from the cumulative total, not per use case.

## Use Case #2: Remote Worker: Zero Trust Access

Foundation · Required

100 pts

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

### STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here, or paste one made with the Cisco Circuit app / an AI tool.

### STEP 2 · ADDRESS THE CUSTOMER'S PAIN POINTS

| CUSTOMER PAIN POINT                                                       | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|---------------------------------------------------------------------------|---------------------------------------------------------|
| Full client VPN has over permissive privileges. Not considered Zero Trust |                                                         |

| CUSTOMER PAIN POINT                                              | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|------------------------------------------------------------------|---------------------------------------------------------|
| User satisfaction is low due to manual VPN connection daily      |                                                         |
| Too many sign-ons per application. Low user satisfaction.        |                                                         |
| Want to eliminate password logins to minimize stolen credentials |                                                         |
| Existing Okta IdP solution getting too expensive                 |                                                         |
| CIO wants to modernize branches with SASE                        |                                                         |

#### STEP 3 · MEET THE CUSTOMER'S REQUIREMENTS

| CUSTOMER REQUIREMENT                                                                                                                      | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| All corporate users can use full client VPN to access selected legacy private applications. But most prefer ZTNA for modern applications. |                                                         |
| All corporate users need controlled internet access (block gambling and gaming)                                                           |                                                         |
| Contractors can only access private applications via clientless RDP access.                                                               |                                                         |
| Improve user experience with minimal application sign-ons                                                                                 |                                                         |
| Customer wants to modernize branches with SASE                                                                                            |                                                         |

#### STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled. Explain **how each works** and **why you'd choose it** (the differentiator). Use the blank row for anything else you'd add.

| CISCO PRODUCT / FEATURE                                        | WHAT IT DOES / HOW IT WORKS | WHY YOU CHOSE IT (CISCO DIFFERENTIATOR) |
|----------------------------------------------------------------|-----------------------------|-----------------------------------------|
| Secure Access, Secure Client, Duo, ZTA module in Secure Client |                             |                                         |
| Secure Client VPN module (Full Tunnel)                         |                             |                                         |
| ZTA module, Secure Client                                      |                             |                                         |
| Secure Access Private Resources proxy                          |                             |                                         |
| Duo                                                            |                             |                                         |
| Secure Access VPNaaS                                           |                             |                                         |
| Secure Access Unified Policy                                   |                             |                                         |
| Secure Access + SD-WAN                                         |                             |                                         |
|                                                                |                             |                                         |

PROCTOR USE ONLY: USE CASE SCORE

| CRITERION                    | MAX | SCORE |
|------------------------------|-----|-------|
| Pains & requirements mapping | 40  |       |
| Cisco products (how & why)   | 30  |       |
| Diagram                      | 30  |       |
| Use case total               | 100 |       |

Carry this use case total to the **Final score & group level** page. The Associate / Professional / Expert level and written feedback are recorded once there, from the cumulative total, not per use case.

Use Case #3: AI Access and Agent Controls

Foundation · Required

100 pts

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here, or paste one made with the Cisco Circuit app / an AI tool.

STEP 2 · ADDRESS THE CUSTOMER’S PAIN POINTS

| CUSTOMER PAIN POINT               | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|-----------------------------------|---------------------------------------------------------|
| Shadow AI usage is an issue today |                                                         |



| CUSTOMER PAIN POINT                                                        | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|----------------------------------------------------------------------------|---------------------------------------------------------|
| Over-privileged MCP servers for applications in the private and public DCs |                                                         |
| Users using personal credentials for ChatGPT login                         |                                                         |
| Downloading of risky AI models                                             |                                                         |

STEP 3 · MEET THE CUSTOMER'S REQUIREMENTS

| CUSTOMER REQUIREMENT                                                                                                      | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Users should not be granted rights to use tools equally. Fine grained control is needed                                   |                                                         |
| Corporate users should only be able to login to their organization’s ChatGPT tenant                                       |                                                         |
| Some remote users will be running Claude Desktop agents. Ensure your design includes controls for those employee systems. |                                                         |
| Some users will be integrating MCP servers to local agents like Claude Code                                               |                                                         |

STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled. Explain **how each works** and **why you’d choose it** (the differentiator). Use the blank row for anything else you’d add.

| CISCO PRODUCT / FEATURE         | WHAT IT DOES / HOW IT WORKS | WHY YOU CHOSE IT (CISCO DIFFERENTIATOR) |
|---------------------------------|-----------------------------|-----------------------------------------|
| AI Access within Secure Access  |                             |                                         |
| AI Gateway within Secure Access |                             |                                         |

| CISCO PRODUCT / FEATURE                     | WHAT IT DOES / HOW IT WORKS | WHY YOU CHOSE IT (CISCO DIFFERENTIATOR) |
|---------------------------------------------|-----------------------------|-----------------------------------------|
| Secure Client with agentic client discovery |                             |                                         |
| Duo IdP                                     |                             |                                         |
|                                             |                             |                                         |

PROCTOR USE ONLY: USE CASE SCORE

| CRITERION                    | MAX | SCORE |
|------------------------------|-----|-------|
| Pains & requirements mapping | 40  |       |
| Cisco products (how & why)   | 30  |       |
| Diagram                      | 30  |       |
| Use case total               | 100 |       |

Carry this use case total to the **Final score & group level** page. The Associate / Professional / Expert level and written feedback are recorded once there, from the cumulative total, not per use case.

Use Case #4: Merger / Acquisition; Extend Segmentation

Bonus · If time permits

100 pts

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here, or paste one made with the Cisco Circuit app / an AI tool.

STEP 2 · ADDRESS THE CUSTOMER’S PAIN POINTS

| CUSTOMER PAIN POINT                                                                                                                               | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Unsure how to extend segmentation to acquired company                                                                                             |                                                         |
| Parent company not ready to add 100 newly acquired staff into their corporate directory. Is there a quick option of using what they already have? |                                                         |
| During transition, minimize disruptive user authentication experience to corporate apps and quickly secure acquired company apps                  |                                                         |
| Acquired company suffered identity breach previously. Still using passwords.                                                                      |                                                         |

STEP 3 · MEET THE CUSTOMER’S REQUIREMENTS

| CUSTOMER REQUIREMENT                                                                                 | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| CISO wants to extend segmentation to acquired company using TrustSec                                 |                                                         |
| Only selected branch users can access a restricted resource at recently acquired company (Site 9951) |                                                         |

STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled. Explain **how each works** and **why you'd choose it** (the differentiator). Use the blank row for anything else you'd add.

| CISCO PRODUCT / FEATURE | WHAT IT DOES / HOW IT WORKS | WHY YOU CHOSE IT (CISCO DIFFERENTIATOR) |
|-------------------------|-----------------------------|-----------------------------------------|
| FMC / FTD               |                             |                                         |
| Duo IdP (Directory)     |                             |                                         |
| Duo IdP (SSO routing)   |                             |                                         |
| SCC                     |                             |                                         |
|                         |                             |                                         |

PROCTOR USE ONLY: USE CASE SCORE

| CRITERION                    | MAX | SCORE |
|------------------------------|-----|-------|
| Pains & requirements mapping | 40  |       |
| Cisco products (how & why)   | 30  |       |
| Diagram                      | 30  |       |
| Use case total               | 100 |       |

Carry this use case total to the **Final score & group level** page. The Associate / Professional / Expert level and written feedback are recorded once there, from the cumulative total, not per use case.

Use Case #5: Zero Trust across multicloud Data Centers

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here, or paste one made with the Cisco Circuit app / an AI tool.

STEP 2 · ADDRESS THE CUSTOMER'S PAIN POINTS

| CUSTOMER PAIN POINT                                                                             | YOUR TEAM'S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Minimal visibility of applications across multicloud data centers                               |                                                         |
| Lacking segmentation strategy and plan                                                          |                                                         |
| Wide deployment of Kubernetes with latency and visibility issues                                |                                                         |
| Unable to patch IoT devices                                                                     |                                                         |
| Too many disparate security management interfaces for on-prem and multicloud enforcement points |                                                         |

STEP 3 · MEET THE CUSTOMER’S REQUIREMENTS

| CUSTOMER REQUIREMENT                                                                                 | YOUR TEAM’S RESPONSE: HOW YOUR CISCO DESIGN SOLVES THIS |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Limited budget. Want to leverage native security features when practical. Open to virtual appliances |                                                         |
| CIO seeking end-to-end segmentation solution                                                         |                                                         |
| Minimize complexity                                                                                  |                                                         |

STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled. Explain **how each works** and **why you’d choose it** (the differentiator). Use the blank row for anything else you’d add.

| CISCO PRODUCT / FEATURE | WHAT IT DOES / HOW IT WORKS | WHY YOU CHOSE IT (CISCO DIFFERENTIATOR) |
|-------------------------|-----------------------------|-----------------------------------------|
| Secure Workload         |                             |                                         |
| NGFW / FTD              |                             |                                         |
| Isovalent               |                             |                                         |
|                         |                             |                                         |

**PROCTOR USE ONLY: USE CASE SCORE**

| CRITERION                    | MAX | SCORE |
|------------------------------|-----|-------|
| Pains & requirements mapping | 40  |       |
| Cisco products (how & why)   | 30  |       |
| Diagram                      | 30  |       |
| Use case total               | 100 |       |

Carry this use case total to the **Final score & group level** page. The Associate / Professional / Expert level and written feedback are recorded once there, from the cumulative total, not per use case.

## Final score & group level

Proctor use. Enter each use case total, add the cumulative totals, then award the group a single level from the rules below.

Team / table #: \_\_\_\_\_ Members: \_\_\_\_\_

| USE CASE                                               | SCORE    |       |
|--------------------------------------------------------|----------|-------|
| Use Case #1: Managed Environment; Segmentation         | Required |       |
| Use Case #2: Remote Worker: Zero Trust Access          | Required |       |
| Use Case #3: AI Access and Agent Controls              | Required |       |
| Use Case #4: Merger / Acquisition; Extend Segmentation | Bonus    |       |
| Use Case #5: Zero Trust across multicloud Data Centers | Bonus    |       |
| Required total (UC1-UC3)                               |          | / 300 |
| Full total (UC1-UC5, if all attempted)                 |          | / 500 |
| Required % (required total ÷ 300 × 100)                |          | %     |
| Full % (full total ÷ 500 × 100)                        |          | %     |

### Award the group level

Associate

All three required use cases completed, cumulative score below 70% (under 210 of 300).

Professional

All three required use cases completed with a cumulative score of 70% or higher (210 of 300 or more). A perfect three-use-case result is Professional.

Expert

All five use cases attempted with a cumulative score of 80% or higher (400 of 500 or more).



### How the level is decided.

- Score every use case out of 100, then add the scores (these are cumulative totals, not per-use-case levels).
- Associate: all three required use cases (UC1-UC3) completed, cumulative under 70% (below 210 of 300).
- Professional: all three required use cases completed, cumulative 70% or higher (210 of 300 or more). A perfect three-use-case result is Professional.
- Expert: all five use cases attempted and cumulative 80% or higher (400 of 500 or more).
- A group that attempts all five but scores under 80% is leveled on the three required use cases (70% or higher is Professional, otherwise Associate).

### FINAL GROUP LEVEL (CIRCLE ONE)

Associate

Professional

Expert

### Feedback for the team

#### What the team did well

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#### Where they can improve

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## Glossary

### **ZTNA** Zero Trust Network Access

Per-application access that replaces broad, over-permissive VPN.

### **SASE** Secure Access Service Edge

Cloud-delivered networking and security combined.

### **SSE** Security Service Edge

The security half of SASE (SWG, ZTNA, CASB, DLP...).

### **SGT** Security Group Tag

An identity label used to segment traffic without VLANs or IP subnets.

### **TrustSec** Cisco TrustSec

Cisco's SGT-based segmentation technology.

### **ISE** Identity Services Engine

Authenticates and profiles users/devices and assigns SGTs.

### **SWG** Secure Web Gateway

Inspects and filters internet-bound web traffic.

### **IdP** Identity Provider

The system that authenticates users (e.g., Duo, Okta, Entra).

### **SSO** Single Sign-On

One login for many applications.

### **MFA** Multi-Factor Authentication

A second proof of identity beyond a password.

### **802.1X** IEEE 802.1X

Port-based network access authentication.

### **MAB** MAC Authentication Bypass

Authenticate a device by its MAC when it can't do 802.1X.

### **pxGrid** Platform Exchange Grid

Cisco's framework for sharing context between security products.

**SGACL** [Security Group ACL](#)

Access rules written in terms of SGTs.

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**CSW** [Cisco Secure Workload](#)

Application visibility and microsegmentation (formerly Tetration).

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**ADM** [Application Dependency Mapping](#)

Discovers app traffic flows to build segmentation policy.

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**FTD / FMC** [Firewall Threat Defense / Firewall Mgmt Center](#)

Cisco Secure Firewall and its manager.

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**NGFW / NGIPS** [Next-Gen Firewall / IPS](#)

Modern firewall and intrusion-prevention.

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**MCP** [Model Context Protocol](#)

How AI agents connect to external tools and data.

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**DCR** [Dynamic Client Registration](#)

Registering and authorizing agents/apps with an IdP.

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**DLP** [Data Loss Prevention](#)

Stops sensitive data from leaving the organization.

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**SCIM** [System for Cross-domain Identity Mgmt](#)

Syncs users between directories (e.g., Okta → Duo).

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**VPNaaS** [VPN as a Service](#)

Cloud-delivered VPN termination.

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**eBPF** [extended Berkeley Packet Filter](#)

Kernel technology for observability and security (Cilium).

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**CNI** [Container Network Interface](#)

Kubernetes networking plugin (e.g., Cilium).

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**SCC** [Security Cloud Control](#)

Unified cloud management for Cisco security policy.

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**IoT** [Internet of Things](#)

Networked devices like printers, cameras and badge readers.

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ZTA Design Clinic Workbook, team answer sheet. Use it alongside the Activity Document.